

CALEB SKINNER

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EDUCATION

Rice University – Houston, TX

- PhD in Statistics (M.S. earned en route) Expected May 2029
- Advisor: Dr. Meng Li
- GPA: 3.97
- Qualifying Exam - Passed Summer 2025

Baylor University – Waco, TX

- Bachelor of Business Administration May 2024
- Majors: Statistics, Economics, Mathematics, Business Fellows
- Honors Program; Overall GPA: 3.97
- Honors Thesis: “Is Momentum Real: On the Existence and Influences of Psychological Momentum in Tennis” (advisor: Dr. Rodney Sturdivant) <https://hdl.handle.net/2104/12634>

PUBLICATIONS AND PREPRINTS

Skinner, C., Guo, Y., & Li, M. (2026). LLM Sparsity Prior for Robust Feature Selection. Under review.
<https://arxiv.org/abs/2605.23102>

- Incorporates LLM-derived feature importance weights into spike-and-slab and spike-and-slab lasso priors with a robust structure for discounting poor weights.

Skinner, C., & Li, M. Exact Partial Constancy Prior for Structured Ensembles. In preparation.

- Develops a Bayesian framework for predictor aggregation across data types, with applications to ensemble learning and forecast aggregation.

Skinner, C., Shi, H., & Luo, H. Optimism for Ensemble Regression. In preparation.

- Derives the expected optimism for least squares ensembles under differing covariance structures. Insights are applicable to several ensemble techniques such as random forests and boosting.

Skinner, C., Patrick, J., Bryson, C., & Sturdivant, R. (2025). Mixed Martial Arts - Interrater Reliability. *The SCORE Network*. <https://doi.org/10.17605/OSF.IO/F8VBQ>

- Interactive teaching tool details Interrater Reliability using data from Mixed Martial Arts judges.

PRESENTATIONS

- ASA Rice Chapter Poster Session – LLM Sparsity Prior for Robust Feature Selection Spring 2026
- Ken Kennedy Institute Retreat – LLM Sparsity Prior for Robust Feature Selection Spring 2026
- Baylor Honors Week – Psychological Momentum in Tennis Spring 2024

RESEARCH PROJECTS

Brain Aneurysm Detection and Localization Fall 2025

- Developed a deep learning pipeline for identification and localization of brain aneurysms from three-dimensional brain scans. Pre-processed multi-modal scans and implemented a two-step 3-D Convolutional Neural Network with modern block architecture.

Sparse Trustworthy Variational Bayes

Spring 2025

- Implemented a novel extension of Trustworthy Variational Bayes (Liu and Li, 2024) applied to the discrete spike-and-slab prior (Ray and Szabo, 2022) to address variance underestimation and improve variational credible intervals coverage.

Sleeper Bayesian

Fall 2024

- Developed a Normal Bayesian Hierarchical Model to estimate NBA player performance and future outcomes. Incorporated game theoretic decision boundaries to optimize fantasy basketball roster decisions.

Baseball Analytics Application, Sports Analytics Club – Vice President

2021 – 2023

- Built and deployed an R Shiny application delivering interactive visualizations and predictive tables utilized by the baseball coaching staff.

TEACHING AND MENTORING

NSF Research Experience for Teachers (RET), Rice University – Graduate Mentor

Summer 2026

- Directed a six-week research project applying statistical inference, regression modeling, and multiple comparison procedures to an acute kidney injury clinical dataset, mentoring a high school mathematics teacher from research question formulation to final poster.

Data to Knowledge Lab Mentor

Spring 2026

- Mentored a team of six undergraduates twice a week through a semester research project applying statistical inference, text clustering, and dashboard development to analyze work orders for a facilities management client, ultimately developing a poster presentation and interactive dashboard.

Rice Teaching Assistant

Fall 2024 – Spring 2026

- Led weekly study sessions and office hours, developed exam questions, and graded assignments for undergraduate and graduate courses.
- STAT 310: Probability and Statistics (2 semesters); STAT 482: Quantitative Financial Analysis; STAT 519: Statistical Inference (graduate).

Mathematical Contest in Modeling (MCM), Problem C Co-Author

Spring 2024

- Co-authored the 2024 MCM Problem C (momentum in tennis) under Dr. Rodney Sturdivant, sourcing and curating the dataset and drafting the data documentation. Problem attempted by over 18,000 teams internationally. <https://www.comap.org/membership/member-resources/item/momentum-in-tennis>

SCORE Network Material Developer

Summer 2023

- Developed three extensive (20-50 page) interactive lesson plans communicating statistical concepts (Cluster Analysis, Interrater Reliability, Logistic Regression).
- Produced interactive and engaging modules utilizing R-Markdown and ISLE.

HONORS AND AWARDS

- Getterman Scholar – Full Ride Scholarship to Baylor University 2020 – 2024
- Crane Scholar (Highly Selective Interdisciplinary Program) 2021 – 2024
- Most Outstanding Undergraduate in Statistics May 2024

TECHNICAL SKILLS

- Languages: R, Python, LaTeX
- Tools: Stan, Shiny, PyTorch, tidyverse, Git, R Markdown/Quarto
- Computing: Linux/bash, remote compute clusters, GPU training

SERVICE

<i>Vice President – Rice ASA Student Chapter</i>	2025 - Present
<i>Judge for Undergraduate Statistics Capstone</i>	Fall 2025, Spring 2026
<i>Prospective Student Preview Day - PhD Coordinator (2026), PhD Representative (2025)</i>	2025 – 2026

PROFESSIONAL MEMBERSHIPS

<i>American Statistical Association</i>	2024 – Present
<i>Sections: Bayesian Statistical Science, Business and Economic Statistics; Houston Chapter</i>	2025 – Present